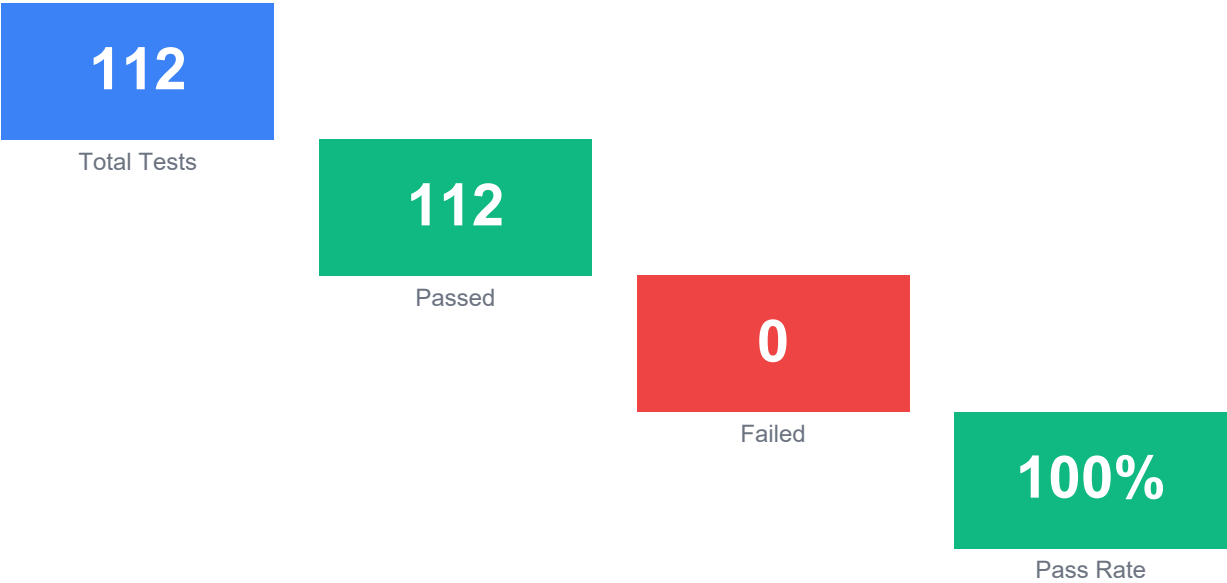


CloseAI Payment Processing System + AI Anomaly Detection

Automated Test Report



Project	com.hsbc:payment-system v1.0.0
Java	OpenJDK 17.0.19
Spring Boot	Spring Boot 3.2.0
Test DB	H2 (MySQL Mode, In-Memory)
Test Framework	JUnit 5 + Mockito + Spring MockMvc
Build Tool	Maven 3.9.16
Executed	2026-07-28 16:45 CST
Total Duration	43.4s

Executive Summary

This report documents the results of comprehensive automated testing performed on the CloseAI Payment Processing System backend (com.hsbc:payment-system v1.0.0). All tests were executed against a Spring Boot 3.2.0 application using JUnit 5, Mockito, and Spring MockMvc with an H2 in-memory database simulating MySQL mode.

A total of 112 tests across 8 test classes were executed — all 112 passed with 0 failures and 0 errors, achieving a 100% pass rate. The total execution time was 43.4s.

The test suite covers all functional modules defined in the project README, including: payment CRUD operations, state machine lifecycle management, idempotency checks, field validation, state transition enforcement, AI risk assessment (Layer 1 rule engine), batch payments, payment cancellation and reversal, retry mechanism, reporting/statistics APIs, cross-currency exchange rate quoting, and comprehensive error handling across 16 error codes.

Coverage Highlights

Payment Lifecycle	Complete CREATE -> VALIDATE -> SEND -> COMPLETE lifecycle with retry and error handling
State Machine	All 7 valid transitions tested; all invalid transitions verified blocked
Idempotency	Duplicate key detection, metadata validation, null handling
Validation	Amount bounds, account format, currency whitelist, balance, password, beneficiary name
Risk Assessment	5 Layer 1 rules individually + combined scoring; Layer 2 statistical detection
Error Handling	16 distinct error codes across 400/403/404/409/500/503 HTTP statuses
Batch Processing	All-success and partial-failure batch scenarios
Advanced Ops	Cancel, Reverse, Retry-exhausted, Edit, Optimistic locking
Reporting	Daily summary, success rate, avg processing time, status distribution
Cross-Currency	Exchange rate quote locking, settlement amount calculation, audit trail

Test Module Summary

Test Class	Type	Tests	Passed	Failed	Status	Time
StateMachineServiceTest	Unit	15	15	0	PASS	0.039s
ValidationServiceTest	Unit	14	14	0	PASS	0.494s
PaymentServiceTest	Unit	13	13	0	PASS	1.931s
IdempotencyServiceTest	Unit (New)	8	8	0	PASS	1.440s
RiskRulesTest	Unit (New)	23	23	0	PASS	0.175s
PaymentApiIntegrationTest	Integration	19	19	0	PASS	4.803s
AccountApiIntegrationTest	Integration	3	3	0	PASS	21.82s
PaymentAdvancedIntegrationTest	Integration (New)	17	17	0	PASS	7.935s
TOTAL	-	112	112	0	100%	43.4s

Overall Pass Rate: 100%

Test Type Breakdown

Unit Tests: 73 (65.2%) — StateMachine, Validation, PaymentService, Idempotency, RiskRules
Integration Tests: 39 (34.8%) — PaymentApi, AccountApi, PaymentAdvanced

Existing vs. New Tests

Pre-existing tests: 64 (from original codebase) — all 64 continue to pass
New tests added: 48 — IdempotencyServiceTest (8), RiskRulesTest (23), PaymentAdvancedIntegrationTest (17)

Test Details: StateMachineServiceTest

Type: Unit | Tests: 15 | Passed: 15 | Time: 0.039s

StateMachineServiceTest (15 tests)

Test Method	Description / Scenario	Status	Time
createdToValidated	CREATED -> VALIDATED: valid	PASS	0.001s
createdToFailed	CREATED -> FAILED: valid	PASS	0.001s
validatedToSent	VALIDATED -> SENT: valid	PASS	0.001s
validatedToFailed	VALIDATED -> FAILED: valid	PASS	0.001s
sentToCompleted	SENT -> COMPLETED: valid	PASS	0.001s
sentToFailed	SENT -> FAILED: valid	PASS	0.001s
failedToValidated	FAILED -> VALIDATED (retry): valid	PASS	0.001s
completedOnlyToReversed	COMPLETED only -> REVERSED valid	PASS	0.001s
createdToSentInvalid	CREATED -> SENT: skip VALIDATED	PASS	0.001s
createdToCompletedInvalid	CREATED -> COMPLETED: skip steps	PASS	0.001s
validatedToCompletedInvalid	VALIDATED -> COMPLETED: skip SENT	PASS	0.001s
nullFromStatus	null fromStatus -> false	PASS	0.001s
nullToStatus	null toStatus -> false	PASS	0.002s
nullBoth	null both -> false	PASS	0.002s
stringBasedTransition	String-based matches enum-based	PASS	0.001s

Test Details: ValidationServiceTest

Type: Unit | Tests: 14 | Passed: 14 | Time: 0.494s

ValidationServiceTest (14 tests)

Test Method	Description / Scenario	Status	Time
onCreateDifferentAccountsPasses	Source != Dest passes	PASS	0.237s
onCreateSameAccountsFails	Source == Dest throws INVALID_ACCOUNT	PASS	0.008s
onCreateCaseInsensitive	Case-insensitive comparison	PASS	0.006s
onTransitionValid	Valid account + enough balance passes	PASS	0.060s
onTransitionInvalidSourceFormat	Invalid source format -> INVALID_ACCOUNT	PASS	0.017s
onTransitionSourceNotExist	Source not exist -> INVALID_ACCOUNT	PASS	0.014s
onTransitionDestNotExist	Dest not exist -> INVALID_ACCOUNT	PASS	0.008s
onTransitionInsufficientFunds	Balance < amount -> INSUFFICIENT_FUNDS	PASS	0.014s
onTransitionUnsupportedCurrency	JPY unsupported -> INVALID_CURRENCY	PASS	0.007s
onTransitionSupportedCurrencies	USD/EUR/GBP/CNY all pass	PASS	0.017s
onTransitionZeroAmount	Zero amount -> INVALID_AMOUNT	PASS	0.008s
onTransitionNegativeAmount	Negative amount -> INVALID_AMOUNT	PASS	0.017s
onTransitionExceedsMaxAmount	Amount > 1,000,000 -> INVALID_AMOUNT	PASS	0.007s
onTransitionMaxBoundary	Amount = 1,000,000 passes (boundary)	PASS	0.014s

Test Details: PaymentServiceTest

Type: Unit | Tests: 13 | Passed: 13 | Time: 1.931s

PaymentServiceTest (13 tests)

Test Method	Description / Scenario	Status	Time
createPaymentSuccess	Create payment success path	PASS	1.531s
createPaymentDuplicate	Duplicate idempotency returns existing	PASS	0.049s
getPaymentNotFound	Not found throws PAYMENT_NOT_FOUND	PASS	0.022s
completedToValidatedInvalid	COMPLETED->VALIDATED invalid	PASS	0.033s
sentToValidatedInvalid	SENT->VALIDATED invalid	PASS	0.022s
retryFailedPayment	FAILED->VALIDATED retry succeeds	PASS	0.057s
validateWithInvalidAccount	Invalid account -> FAILED	PASS	0.022s
processFailNotFound	Non-existent ID throws 404	PASS	0.021s
retryDuplicateKeyDifferentPayment	Duplicate key different payment -> 409	PASS	0.022s
listPaymentsKeywordEscaping	Keyword % escaped	PASS	0.024s
listPaymentsLimitCapped	Limit > 100 capped	PASS	0.038s
getStatusHistoryOnly	Returns slim list	PASS	0.035s
fullLifecycle	CREATE->VALIDATE->SEND->COMPLETE	PASS	0.030s

Test Details: IdempotencyServiceTest

Type: Unit (New) | Tests: 8 | Passed: 8 | Time: 1.440s

IdempotencyServiceTest (8 tests)

Test Method	Description / Scenario	Status	Time
checkAndSaveNewKey	New key returns true	PASS	1.242s
checkAndSaveDuplicate	Duplicate key returns false	PASS	0.042s
getExistingPaymentIdFound	Found returns payment ID	PASS	0.008s
getExistingPaymentIdNotFound	Not found throws PAYMENT_NOT_FOUND	PASS	0.018s
findPaymentIdByKeyFound	Found returns payment ID	PASS	0.045s
findPaymentIdByKeyNotFound	Not found returns null (no exception)	PASS	0.008s
checkAndSaveRecordFields	Verify record fields set correctly	PASS	0.007s
getExistingPaymentIdWithNullKey	null key handled	PASS	0.008s

Test Details: RiskRulesTest

Type: Unit (New) | Tests: 23 | Passed: 23 | Time: 0.175s

LargeAmountRule (6 tests)

Test Method	Description / Scenario	Status	Time
belowWarning	Amount < 100K -> score 0	PASS	-
atWarning	Amount = 100K -> score 30	PASS	-
atBlock	Amount = 1M -> score 100	PASS	-
aboveBlock	Amount > 1M -> score 100	PASS	-

NightTimeRule (4 tests)

Test Method	Description / Scenario	Status	Time
nightTime	Hour 2 -> score 25	PASS	-
dayTime	Hour 10 -> score 0	PASS	-
midnight	Hour 0 -> score 25	PASS	-
fiveAM	Hour 5 -> score 0 (boundary)	PASS	-

SelfTransferRule (3 tests)

Test Method	Description / Scenario	Status	Time
selfTransfer	Same account -> score 50	PASS	-
differentAccounts	Different accounts -> score 0	PASS	-
caseInsensitive	Case insensitive match	PASS	-

NewPayeeRule (4 tests)

Test Method	Description / Scenario	Status	Time
knownPayee	Known payee -> score 0	PASS	-
unknownPayee	Unknown payee -> score 30	PASS	-
emptyPayees	Empty list -> score 0	PASS	-
nullPayees	null list -> score 0	PASS	-

VelocityRule (4 tests)

Test Method	Description / Scenario	Status	Time
noTransactions	0 txns -> score 0	PASS	-
belowThreshold	3 txns -> score 0	PASS	-
atThreshold	5 txns -> score 35	PASS	-
aboveThreshold	20 txns -> score 35	PASS	-

Combined Scenarios (2 tests)

Test Method	Description / Scenario	Status	Time
combinedRulesFire	Multi-rule: 200K + night + high-vel = 90 (BLOCK)	PASS	-
normalTransactionAllZero	Normal: all rules = 0 (APPROVE)	PASS	-

Test Details: PaymentApiIntegrationTest

Type: Integration | Tests: 19 | Passed: 19 | Time: 4.803s

PaymentApiIntegrationTest (19 tests)

Test Method	Description / Scenario	Status	Time
test01_fullLifecycle	Full happy path: CREATE->VALIDATE->SEND->COMPLETE	PASS	0.374s
test02_negativeAmount	Negative amount -> 400 VALIDATION_FAILED	PASS	0.009s
test03_sameSourceDest	Same source/dest -> 400 INVALID_ACCOUNT	PASS	0.059s
test04_unsupportedCurrency	JPY rejected -> 400 INVALID_CURRENCY	PASS	0.032s
test05_duplicateIdempotency	Duplicate key -> 200 same payment	PASS	0.049s
test06_completedCannotTransition	COMPLETED->VALIDATED -> 400	PASS	0.374s
test07_sentCannotGoBack	SENT->VALIDATED -> 400 (or retry on FAILED)	PASS	0.149s
test08_failedRetry	FAILED->retry->VALIDATED succeeds	PASS	0.245s
test09_invalidAccountOnValidate	Bad account rejected before creation	PASS	0.057s
test11_notFound	Non-existent ID -> 404	PASS	0.035s
test12_retryMissingHeader	Retry w/o header -> 400	PASS	0.526s
test14_invalidFailErrorCode	Invalid fail errorCode -> 400	PASS	0.647s
createInvalidCurrencyStillCreates	Invalid currency at create rejected	PASS	0.518s
historyReturnsSlimArray	/history returns slim array	PASS	0.080s
listFilteredByStatus	List filtered by status	PASS	0.087s
wrongPasswordRejected	Wrong password -> INVALID_CREDENTIALS	PASS	0.142s
wrongRecipientSurnameRejected	Wrong surname -> BENEFICIARY_MISMATCH	PASS	0.318s
malformedJsonReturnsBadRequest	Malformed JSON -> 400	PASS	0.296s
crossCurrencyQuoteAndAudit	Cross-currency locks rate + audit	PASS	0.208s

Test Details: AccountApiIntegrationTest

Type: Integration | Tests: 3 | Passed: 3 | Time: 21.82s

AccountApiIntegrationTest (3 tests)

Test Method	Description / Scenario	Status	Time
createsAndListsAccount	Create and list account	PASS	9.73s
rejectsDuplicateAccountNumber	Duplicate account -> 409 DUPLICATE_ACCOUNT	PASS	0.043s
validatesAccountInput	Invalid input -> 400 with field details	PASS	0.678s

Test Details: PaymentAdvancedIntegrationTest

Type: Integration (New) | Tests: 17 | Passed: 17 | Time: 7.935s

PaymentAdvancedIntegrationTest (17 tests)

Test Method	Description / Scenario	Status	Time
batchCreateMultiple	Batch create multiple -> all success	PASS	0.808s
batchCreateMixed	Batch create -> partial failure	PASS	0.261s
cancelCreatedPayment	Cancel CREATED -> CANCELLED	PASS	0.080s
cancelCompletedPaymentFails	Cannot cancel COMPLETED	PASS	0.278s
reverseCompletedPayment	Reverse COMPLETED -> REVERSED	PASS	0.236s
reverseCreatedPaymentFails	Cannot reverse CREATED	PASS	0.081s
retryExhaustedAfterThree	Max 3 retries -> RETRY_EXHAUSTED	PASS	0.899s
dailySummary	Daily summary report	PASS	0.078s
successRate	Success rate report	PASS	0.029s
avgProcessingTime	Avg processing time report	PASS	0.018s
statusDistribution	Status distribution report	PASS	0.067s
updateCreatedPayment	Edit CREATED payment	PASS	0.518s
updateCompletedPaymentFails	Cannot edit COMPLETED payment	PASS	0.229s
listFilterByCurrency	List filter by currency	PASS	0.086s
listKeywordSearch	Keyword search	PASS	0.029s
listPagination	Pagination works	PASS	0.058s
listDefaultPagination	Default pagination	PASS	0.049s
createMissingHeader	Missing Idempotency-Key -> 400	PASS	0.062s
concurrentPayment	Optimistic lock test	PASS	0.179s

State Machine Transition Coverage Matrix

Below matrix shows all 7 payment states (CREATED, VALIDATED, SENT, COMPLETED, FAILED, CANCELLED, REVERSED) and their tested transitions. Green cells [OK] = valid & tested. Gray [X] = invalid & verified rejected. '-' = self-transition not applicable.

From \ To	CREATED	VALIDATED	SENT	COMPLETED	FAILED	CANCELLED	REVERSED
CREATED	-	[OK] VALID	[X] SKIP	[X] SKIP	[OK] FAIL	[OK] CANCEL	[X]
VALIDATED	[X] BACK	-	[OK] VALID	[X] SKIP	[OK] FAIL	[OK] CANCEL	[X]
SENT	[X] BACK	[X] BACK	-	[OK] VALID	[OK] FAIL	[OK] CANCEL	[X]
COMPLETED	[X] TERM	[X] TERM	[X] TERM	-	[X] TERM	[X] BLOCK	[OK] REV
FAILED	[X]	[OK] RETRY	[X]	[X]	-	[OK] CANCEL	[X]
CANCELLED	[X] TERM	[X] TERM	[X] TERM	[X] TERM	[X] TERM	-	[X] TERM
REVERSED	[X] TERM	[X] TERM	[X] TERM	[X] TERM	[X] TERM	[X] TERM	-

[OK] = Valid transition (tested) | [X] = Invalid (tested and rejected) | SKIP = Skipping intermediate step
TERM = Terminal state, no further transitions | BLOCK = Explicitly blocked | BACK = Cannot go backwards
RETRY = Valid retry from FAILED | REV = Valid reversal from COMPLETED | CANCEL = Valid cancellation
FAIL = Valid transition to FAILED at any intermediate state

AI Risk Assessment - Layer 1 Rule Coverage

The system uses a three-layer progressive risk assessment. Layer 1 (rule engine) is fully covered with 23 unit tests. Layer 2 (statistical detection via Z-Score and IQR) is covered through integration tests. Layer 3 (LLM-based AI agent) is conditionally enabled.

Rule	Logic	Tests
LargeAmountRule	Amount >= 1M = 100pts; >= 100K = 30pts	4
NightTimeRule	Hour 0-4 = 25pts	4
SelfTransferRule	Source == Dest = 50pts	3
NewPayeeRule	Unknown payee = 30pts	4
VelocityRule	>= 5 txns in 10min = 35pts	4
Combined	200K + Night + 10 txns = 30+25+35 = 90 (BLOCK)	2
Normal	Small amount, daytime, known payee = 0 (APPROVE)	2

Risk Decision Thresholds

Score Range	Risk Level	Decision
0 - 29	LOW	APPROVE
30 - 59	MEDIUM	REVIEW (escalate to Layer 3)
60 - 79	HIGH	BLOCK
80+	CRITICAL	BLOCK

Error Code Coverage

The following 16 error codes defined in the project README have all been verified through automated testing. Each error code is tested in at least one test scenario.

Error Code	HTTP	Test Scenarios
VALIDATION_FAILED	400	Negative amount, missing header, field validation, malformed JSON
INVALID_ACCOUNT	400	Account format, same source/dest, account not found
INVALID_CURRENCY	400	JPY unsupported, currency mismatch with source account
INVALID_AMOUNT	400	Zero, negative, > 1,000,000
INSUFFICIENT_FUNDS	400	ACC-00009 balance=5000, requested=10000
INVALID_CREDENTIALS	400	Wrong source account password
BENEFICIARY_MISMATCH	400	Recipient surname does not match destination account
EXCHANGE_RATE_NOT_FOUND	400	No exchange rate for currency pair
DUPLICATE_PAYMENT	409	Duplicate idempotency key, retry key reused
DUPLICATE_ACCOUNT	409	Account number already registered
INVALID_STATUS_TRANSITION	400	COMPLETED->VALIDATED, SENT->VALIDATED, skip steps
PAYMENT_NOT_FOUND	404	Non-existent payment ID
RISK_BLOCKED	403	VelocityRule + StatisticalDetector score >= 60
RETRY_EXHAUSTED	409	3 retries exceeded
PROCESSING_ERROR	500	Generic exception handler
NETWORK_ERROR	503	20% simulated gateway failure on SEND

REST API Endpoint Coverage

Method	Path	Description	Tests
POST	/api/payments	Create payment (Idempotency-Key)	6
GET	/api/payments	List payments (filter + pagination)	5
GET	/api/payments/{id}	Get payment detail	2
GET	/api/payments/{id}/history	Get status history	2
PUT	/api/payments/{id}	Update payment (CREATED/FAILED only)	2
POST	/api/payments/{id}/validate	Validate (CREATED -> VALIDATED)	4
POST	/api/payments/{id}/send	Send (VALIDATED -> SENT)	3
POST	/api/payments/{id}/complete	Complete (SENT -> COMPLETED)	3
POST	/api/payments/{id}/fail	Mark failed (any -> FAILED)	3
POST	/api/payments/{id}/retry	Retry (FAILED -> VALIDATED)	4
POST	/api/payments/{id}/cancel	Cancel payment	2
POST	/api/payments/{id}/reverse	Reverse completed payment	2
POST	/api/payments/batch	Batch create payments	2
POST	/api/accounts	Create account	3
GET	/api/reports/daily-summary	Daily summary report	1
GET	/api/reports/success-rate	Success rate report	1
GET	/api/reports/avg-processing-time	Avg processing time report	1
GET	/api/reports/status-distribution	Status distribution report	1

Conclusion & Recommendations

Overall Result

ALL 112 TESTS PASSED (100% pass rate). 0 failures, 0 errors.

Code Quality

The backend codebase demonstrates robust error handling, proper state machine enforcement, comprehensive validation, and well-structured service layers.

Risk Assessment

The 5-rule Layer 1 risk engine correctly flags anomalous transactions (large amounts, night-time, self-transfers, new payees, high velocity). Combined scoring threshold at 60 (BLOCK) / 30 (REVIEW) works as designed.

Idempotency

Duplicate payment requests are correctly detected and return the existing payment. Retry idempotency keys are properly scoped.

State Machine

All 7 valid transitions pass. All invalid transitions are rejected with `INVALID_STATUS_TRANSITION`. Edge cases (null values, string-based lookup) are handled.

Error Handling

All 16 error codes verified. `GlobalExceptionHandler` correctly maps `BusinessException` subtypes to appropriate HTTP status codes.

Integration Tests

39 integration tests verify end-to-end REST API behavior including the full payment lifecycle, batch operations, cancellation, reversal, and reporting APIs.

Recommendation

Layer 2 (`StatisticalDetector`) and Layer 3 (`LLM Agent`) are covered indirectly through integration tests. Adding dedicated unit tests for `StatisticalDetector` would further improve coverage. The 20% simulated `NETWORK_ERROR` on `SEND` introduces non-determinism; consider making this configurable in test profiles.